

Base HalluGuard vs. Router-Augmented HalluGuard

MSE delta is measured against the uncorrected forecast; lower values are better.

Evaluation item	Version 1: Base HalluGuard	Version 2: + Router
Mechanism	Fixed HalluGuard correction calibrated on validation data	Validation-only router selects the correction action per sample
Clean mean MSE delta	-0.623%	-1.289%
Improved clean configs	15 / 16	15 / 16
Control evidence	Beats random controls 15 / 16; paired win rate 0.9375	Beats random action 15 / 16; paired win rate 0.9500
Stress evidence	Boundary stress: -0.932%	Stress mean: -1.391%; boundary stress: -1.483%
Action space	Single calibrated repair policy	No correction, boundary repair, dynamics repair, and smoothing actions
Main finding	Boundary-aware local dynamics gives a stable post-processing signal	Routing improves the correction by matching actions to forecast shapes
Main limitation	Fixed correction cannot adapt to heterogeneous forecast errors	External PatchTST harm remains visible in the compact diagnostic fixture

Source: local prototype experiments summarized in docs/research_narrative.zh-CN.md. The table reports preliminary evidence, not final benchmark claims.